



MAINCRAFTS TECHNOLOGY — CYBERSECURITY & ETHICAL HACKING

Build Your Personal Cybersecurity Lab

Task 2 — Full Lab Build Report

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Task	Task-2: Personal Cybersecurity Lab Build
Environment	VirtualBox • Kali Linux • OWASP Juice Shop
Report type	Full Lab Build Report + Evidence

1 Objective

The objective of this task was to build a safe, fully isolated cybersecurity practice environment on a personal computer. This lab forms the foundation for future exercises in vulnerability assessment, web application testing, exploitation practice, and incident response. The lab consists of an attacker machine (Kali Linux) and a vulnerable target application (OWASP Juice Shop), connected through a properly segmented virtual network.

Scope of this build

- 1 × Attacker machine — Kali Linux (VirtualBox VM)
- 1 × Vulnerable target application — OWASP Juice Shop
- Network segmentation using NAT (internet access) + Host-Only (isolated lab traffic)
- Validation using ping, Nmap, Burp Suite, and Wireshark

Skills demonstrated

Skill Area	Practical Outcome
Virtualization	Running multiple isolated systems for pentesting
Lab isolation	Preventing real-world system damage
Web target setup	Deploying an intentionally vulnerable application
Basic networking	NAT vs Host-Only adapters and subnet usage
Tool awareness	Nmap, Burp Suite, and Wireshark validation

2 Lab Architecture

The lab runs entirely inside VirtualBox on the host machine. The Kali Linux attacker VM is configured with two virtual network adapters: one on NAT (for internet access and software updates) and one on a Host-Only network (for isolated, host-to-VM lab traffic). The target web application is reachable over the Host-Only network, keeping all attack traffic contained and never exposed to the public internet.

Component	Role	Network
Host Computer	Runs VirtualBox hypervisor	Host-Only gateway 192.168.56.1
Kali Linux VM	Attacker machine	NAT: 10.0.2.15 Host-Only: 192.168.56.101
OWASP Juice Shop	Vulnerable target application	192.168.56.1 : 3000

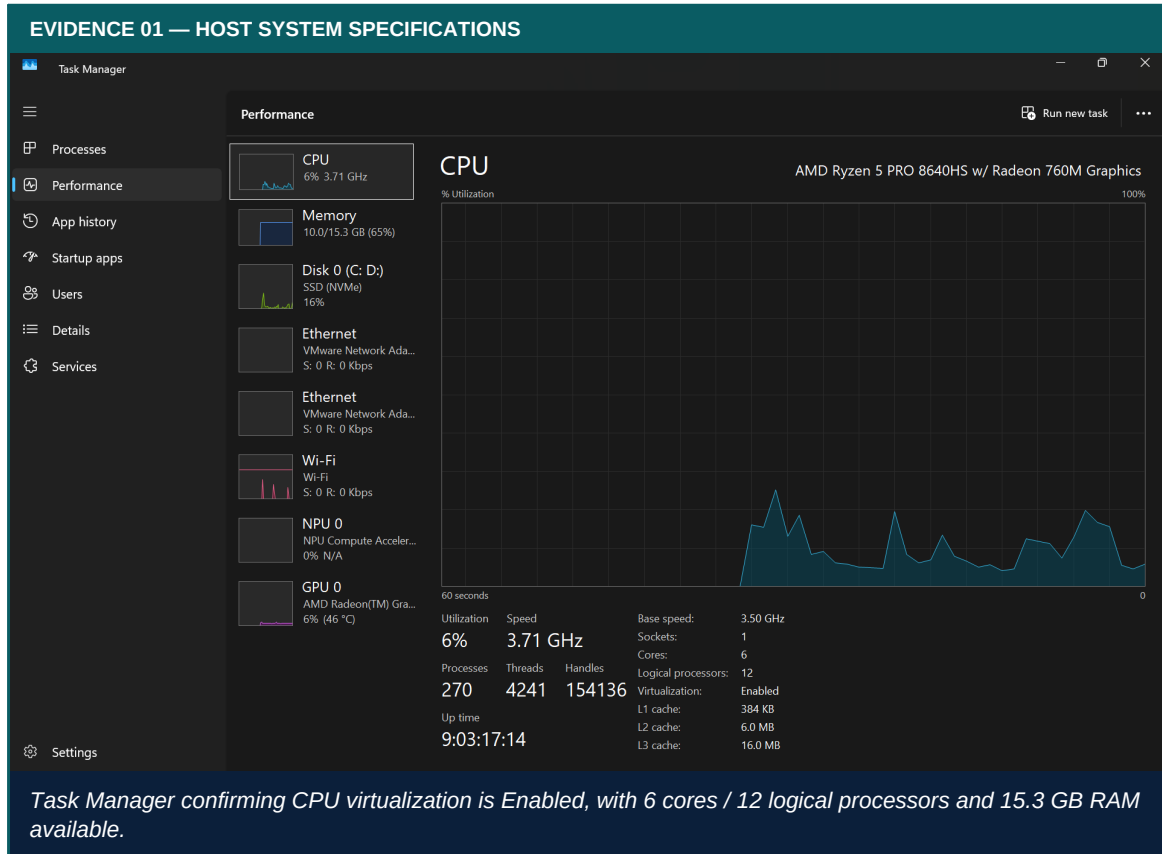
Network purpose

Network	Purpose
NAT (Adapter 1)	Provides internet access to the Kali VM for updates and tool installation
Host-Only (Adapter 2)	Isolated subnet (192.168.56.0/24) used purely for lab traffic between Kali and the target — never reaches the public internet

3 Environment Setup

3.1 Host System & Virtualization Check

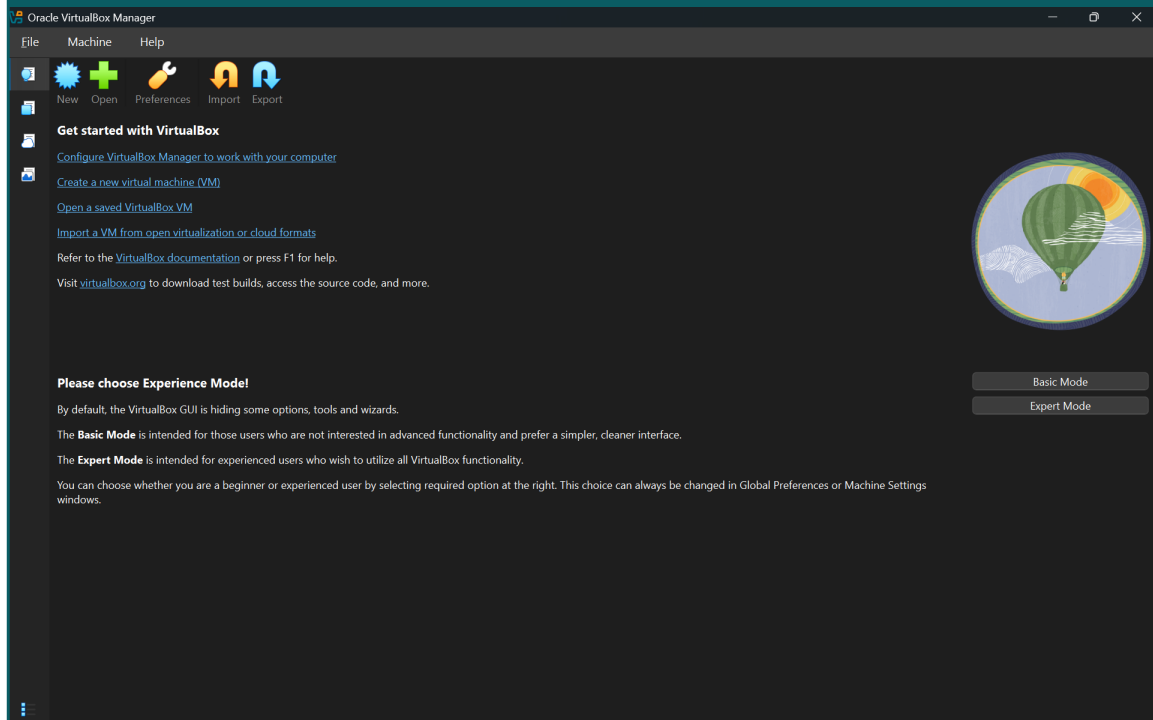
Before installing VirtualBox, the host system was confirmed to have virtualization enabled at the CPU level and sufficient resources (RAM, CPU cores) to run multiple VMs simultaneously.



3.2 VirtualBox Installed & Ready

Oracle VirtualBox Manager was installed successfully and launched, confirming the hypervisor is ready to create and manage virtual machines.

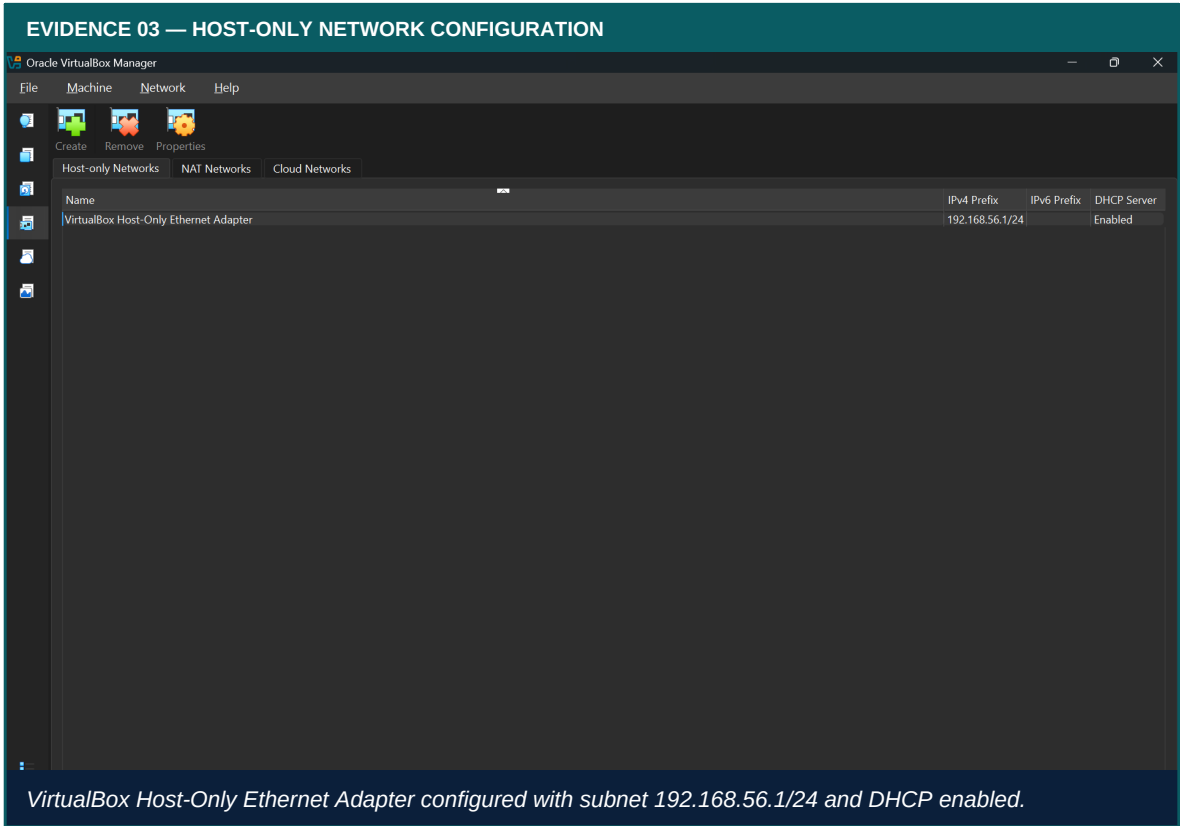
EVIDENCE 02 — VIRTUALBOX INSTALLATION



Oracle VirtualBox Manager home screen after a successful installation.

3.3 Host-Only Network Created

A dedicated Host-Only network was created in VirtualBox to carry isolated lab traffic, with DHCP enabled and a gateway address of 192.168.56.1/24.



3.4 Kali Linux VM — Specifications & Adapters

The Kali Linux attacker VM was deployed with the following specifications and dual network adapters, matching the architecture defined above.

Property	Value
VM Name	kali-linux-2026.2-virtualbox-amd64
Operating System	Debian (64-bit) — Kali Rolling 2026.2
Base Memory	4096 MB
Processors	2 vCPUs
Storage	80.09 GB (SATA)
Adapter 1	NAT
Adapter 2	Host-Only Adapter (VirtualBox Host-Only Ethernet Adapter)

EVIDENCE 04 — KALI VM SPECIFICATIONS

The screenshot shows the VirtualBox VM details pane for a Kali Linux VM. The VM is named 'kali-linux-2026.2-virtualbox-amd64' and is currently 'Powered Off'. The details are organized into several sections:

- System:** Operating System: Debian (64-bit); Base Memory: 4096 MB; Processors: 2; Boot Order: Hard Disk, Optical; Acceleration: Nested Paging, PAE/NX, KVM Paravirtualization.
- Display:** Video Memory: 128 MB; Graphics Controller: VMSVGA; Remote Desktop Server: Disabled; Recording: Disabled.
- Storage:** Controller: IDE; IDE Secondary Device 0: [Optical Drive] Empty; Controller: SATA; SATA Port 0: kali-linux-2026.2-virtualbox-amd64.vdi (Normal, 80.09 GB).
- Audio:** Host Driver: Windows DirectSound; Controller: ICH AC97.
- Network:** Adapter 1: Intel PRO/1000 MT Desktop (NAT); Adapter 2: Intel PRO/1000 MT Desktop (Host-only Adapter, 'VirtualBox Host-Only Ethernet Adapter').
- USB:** USB Controller: OHCI; Device Filters: 0 (0 active).
- Shared folders:** None.
- Description:** Kali Rolling (2026.2) x64; 2026-06-16.

A callout box on the right side of the details pane contains the text: **kali-linux-2026.2-virtualbox-amd64**.

Kali Linux VM details pane showing system resources and both network adapters (NAT + Host-Only).

4 IP Configuration

From inside the Kali VM, the network interfaces were inspected using `ip addr` and `ip route` to confirm both adapters received valid addresses and that routing was correctly split between the NAT and Host-Only networks.

Interface	Network	IP Address
eth0	NAT	10.0.2.15/24
eth1	Host-Only	192.168.56.101/24
Default route	via eth0	10.0.2.2

EVIDENCE 05 — IP CONFIGURATION

```
Applications  ons Edit View Help
kali@kali: ~
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:12:47:00 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 86346sec preferred_lft 86346sec
   inet6 fd37:020c:0377:2:5721:1bf4:2201:acc9/64 scope global dynamic noprefixroute
       valid_lft 86347sec preferred_lft 14347sec
   inet6 fe80::180a:6207:998d:eb0e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:c2:47:00 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
       valid_lft 546sec preferred_lft 546sec
   inet6 fe80::b403:71bc:a94:28ba/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
kali@kali: ~
$ ip route
default via 10.0.2.2 dev eth0 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev eth0 proto kernel scope link src 10.0.2.15 metric 100
192.168.56.0/24 dev eth1 proto kernel scope link src 192.168.56.101 metric 101
kali@kali: ~
```

Output of 'ip addr' and 'ip route' on Kali, confirming eth0 (NAT: 10.0.2.15) and eth1 (Host-Only: 192.168.56.101).

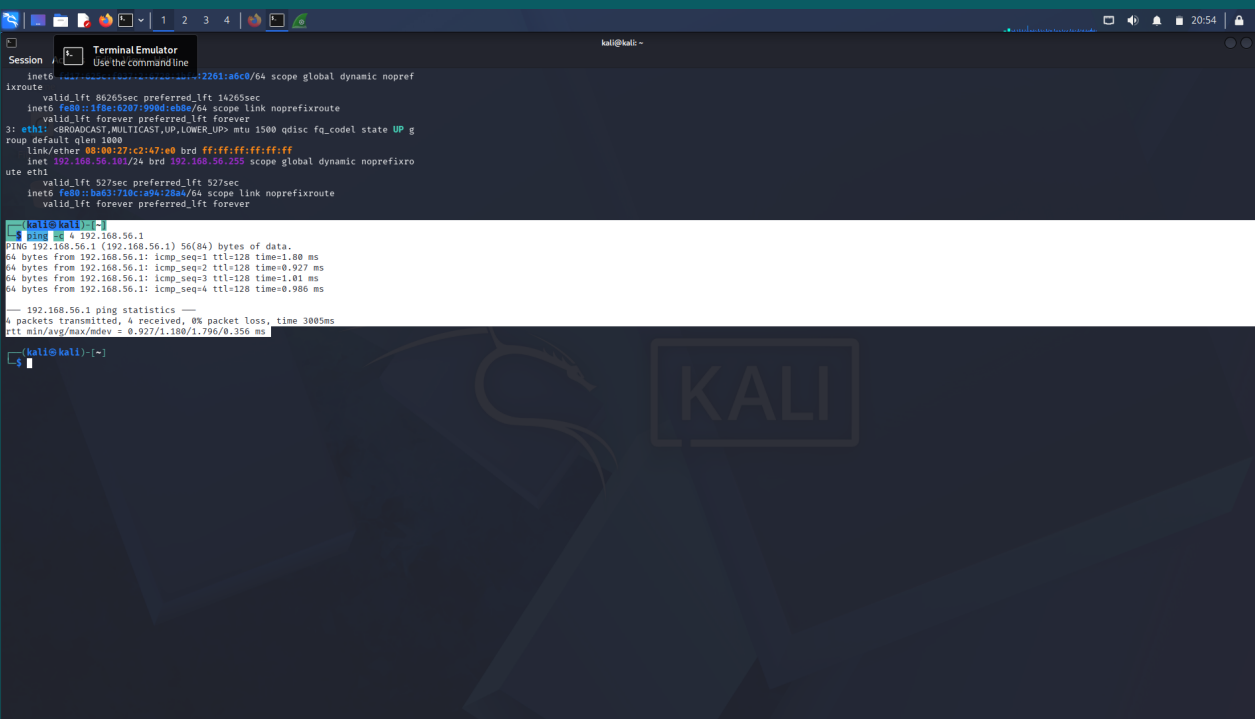
5 Ping & Nmap Validation

Connectivity between the Kali attacker VM and the target over the Host-Only network was validated using ICMP ping, and the target's open service was confirmed using an Nmap port scan.

5.1 Ping Validation

A ping test from Kali to the Host-Only gateway (192.168.56.1) completed with **0% packet loss**, confirming the Host-Only network path is active.

EVIDENCE 06 — PING VALIDATION



```
Session / Terminal Emulator Use the command line kali@kali -
inet6 fe80::12261:a6c0/64 scope global dynamic nopref
ixroute
    valid_lft 86265sec preferred_lft 14265sec
inet6 fe80::1f8e:6207:99ad:eb0e/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 08:00:27:c2:47:e0 brd ff:ff:ff:ff:ff:ff
inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixro
ute eth1
    valid_lft 527sec preferred_lft 527sec
inet6 fe80::ba83:7128:c994:288e/64 scope link noprefixroute
    valid_lft forever preferred_lft forever

kali@kali:~$ ping -c 4 192.168.56.1
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data:
64 bytes from 192.168.56.1: icmp_seq=1 ttl=128 time=1.80 ms
64 bytes from 192.168.56.1: icmp_seq=2 ttl=128 time=0.927 ms
64 bytes from 192.168.56.1: icmp_seq=3 ttl=128 time=1.01 ms
64 bytes from 192.168.56.1: icmp_seq=4 ttl=128 time=0.986 ms

--- 192.168.56.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.927/1.180/1.796/0.356 ms

kali@kali:~$
```

ping -c 4 192.168.56.1 from Kali — 4 packets transmitted, 4 received, 0% packet loss.

5.2 Nmap Port Scan

An Nmap scan against port 3000 on the target confirmed the OWASP Juice Shop web service is reachable and listening over the Host-Only network.

Target	Port	State	Service
192.168.56.1	3000/tcp	open	Juice Shop (Node.js app)

EVIDENCE 07 — NMAP VALIDATION



6 Burp Suite Interception

Burp Suite was configured as an intercepting proxy in front of the browser used to access OWASP Juice Shop. With intercept switched on, a live HTTP request was captured mid-flight, confirming the proxy is correctly positioned between the browser and the target application.

EVIDENCE 08 — BURP SUITE INTERCEPTION

The image displays a side-by-side comparison of a web browser window and the Burp Suite application. On the right, the browser window shows the OWASP Juice Shop website with a grid of juice products: Apple Juice (1000ml) for 1.99€, Apple Pomace for 0.89€, Banana Juice (1000ml) for 1.99€, Basil Smoothie for 2.99€, and Berry Juice (1000ml) for 3.49€. A notification banner at the bottom right of the browser says 'This website uses fruit cookies to ensure you get the juiciest tracking experience. But no real!'. On the left, the Burp Suite interface shows the 'Proxy' tab with 'Intercept on' selected. The 'HTTP history' pane shows a list of requests, with the first one selected. The 'Request' pane shows the details of the selected GET request to http://192.168.56.1:3000/. The 'Inspector' pane shows the 'Selected text' as 'GET / HTTP/1.1' and the 'Decoded from' as 'Host: 192.168.56.1:3000'. The 'Request attributes' pane shows 'Request query parameters' as 0, 'Request body parameters' as 0, 'Request cookies' as 0, and 'Request headers' as 7.

Burp Suite Proxy > Intercept capturing a live GET / request to http://192.168.56.1:3000 from the Juice Shop browser session.

7 Wireshark Packet Capture

A packet capture was taken on the Host-Only interface (eth1) while browsing the target application. The capture shows HTTP, TCP, and WebSocket traffic flowing between the Kali VM (192.168.56.101) and the target (192.168.56.1), confirming full visibility into lab traffic for future analysis tasks.

EVIDENCE 09 — WIRESHARK PACKET CAPTURE

Capturing from eth1

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-F>

No.	Time	Source	Destination	Protocol	Length	Info
53	17.376557933	192.168.56.1	192.168.56.101	HTTP	456	HTTP/1.1 304 Not Modified
58	17.392665917	192.168.56.1	192.168.56.101	HTTP	459	HTTP/1.1 304 Not Modified
59	17.392665819	192.168.56.1	192.168.56.101	HTTP	457	HTTP/1.1 304 Not Modified
61	17.393637812	192.168.56.1	192.168.56.101	HTTP	459	HTTP/1.1 304 Not Modified
158	18.205932403	192.168.56.1	192.168.56.101	HTTP	371	HTTP/1.1 304 Not Modified
125	17.921899983	192.168.56.101	192.168.56.1	HTTP	547	POST /socket.io/?EIO=4&transport=polling&t=Q18IDhe&sid=LpdgiohOWw5TctZtoAAAI HTTP/1.1 (text/plain)
143	17.976310841	192.168.56.1	192.168.56.101	WebSoc...	74	WebSocket Text [FIN]
142	17.973662252	192.168.56.101	192.168.56.1	WebSoc...	78	WebSocket Text [FIN] [MASKED]
148	18.002521495	192.168.56.101	192.168.56.1	WebSoc...	73	WebSocket Text [FIN] [MASKED]
266	33.081194134	192.168.56.101	192.168.56.1	TCP	78	[TCP Keep-Alive ACK] 56904 → 3800 [ACK] Seq=573 Ack=138 Win=64512 Len=0 TSval=1854471997 TSecr=51475967 SLE=137 SRE=138
265	33.081166527	192.168.56.1	192.168.56.101	TCP	60	[TCP Keep-Alive] 3000 → 56904 [ACK] Seq=137 Ack=573 Win=64768 Len=1
269	42.561447760	192.168.56.101	192.168.56.1	TCP	1244	[TCP Keep-Alive] 3000 → 3000 [ACK] Seq=13710 Ack=580 Win=64512 Len=0 TSval=51476194 TSecr=50937880 [TCP RST reassemb...
267	42.562835872	192.168.56.1	192.168.56.101	WebSoc...	69	WebSocket Text [FIN]
268	42.562368254	192.168.56.101	192.168.56.1	TCP	66	56904 → 3000 [ACK] Seq=573 Ack=141 Win=64512 Len=0 TSval=1854481478 TSecr=51508402
269	42.560624679	192.168.56.101	192.168.56.1	WebSoc...	73	WebSocket Text [FIN] [MASKED]
270	42.607156905	192.168.56.1	192.168.56.101	TCP	66	3000 → 56904 [ACK] Seq=141 Ack=580 Win=64768 Len=0 TSval=51508447 TSecr=1854481482
271	57.574889367	192.168.56.1	192.168.56.101	TCP	60	[TCP Keep-Alive] 3000 → 56904 [ACK] Seq=140 Ack=580 Win=64768 Len=1
272	57.574941394	192.168.56.101	192.168.56.1	TCP	78	[TCP Keep-Alive ACK] 56904 → 3000 [ACK] Seq=580 Ack=141 Win=64512 Len=0 TSval=1854496491 TSecr=51508447 SLE=140 SRE=141
273	67.580971676	192.168.56.1	192.168.56.101	WebSoc...	69	WebSocket Text [FIN]
274	67.581825946	192.168.56.101	192.168.56.1	TCP	66	56904 → 3000 [ACK] Seq=580 Ack=144 Win=64512 Len=0 TSval=1854506497 TSecr=51525409
275	67.584895228	192.168.56.101	192.168.56.1	WebSoc...	73	WebSocket Text [FIN] [MASKED]
276	67.624013817	192.168.56.1	192.168.56.101	TCP	66	3000 → 56904 [ACK] Seq=144 Ack=587 Win=64768 Len=0 TSval=51525452 TSecr=1854506500
277	82.591606689	192.168.56.1	192.168.56.101	TCP	60	[TCP Keep-Alive] 3000 → 56904 [ACK] Seq=143 Ack=587 Win=64768 Len=1
278	82.591649073	192.168.56.101	192.168.56.1	TCP	78	[TCP Keep-Alive ACK] 56904 → 3000 [ACK] Seq=587 Ack=144 Win=64512 Len=0 TSval=1854521507 TSecr=51525452 SLE=143 SRE=144

Frame 254: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface eth1, id 0

Ethernet II, Src: 0a:00:27:00:00:18 (0a:00:27:00:00:18), Dst: PCSysteMtec.c2:47:e0 (08:00:27:c2:47:e0)

Internet Protocol Version 4, Src: 192.168.56.1, Dst: 192.168.56.101

Transmission Control Protocol, Src Port: 3000, Dst Port: 56904, Seq: 67445, Ack: 2394, Len: 0

eth1: <live capture in progress>

Packets: 278

Profile: Default

Wireshark capturing live traffic on interface eth1 between 192.168.56.1 and 192.168.56.101, including HTTP and WebSocket sessions.

8 Deliverables Checklist

The table below maps each requested deliverable to the evidence produced in this report.

Deliverable	Evidence Provided	Status
VM list	Kali VM specifications (Evidence 04)	✓ Complete
IP configuration	ip addr / ip route output (Evidence 05)	✓ Complete
Ping / Nmap validation	Ping + Nmap scan results (Evidence 06 & 07)	✓ Complete
Burp screenshot	Intercepted HTTP request (Evidence 08)	✓ Complete
Wireshark screenshot	Live packet capture (Evidence 09)	✓ Complete

9 Conclusion

This task established a fully functional, isolated cybersecurity lab consisting of a Kali Linux attacker machine and an OWASP Juice Shop target, connected through a properly segmented NAT + Host-Only network. Connectivity and tooling were validated end-to-end: ICMP reachability, an Nmap port scan confirming the target service, live HTTP interception in Burp Suite, and a full packet capture in Wireshark. This environment provides a safe, repeatable foundation for the upcoming vulnerability assessment, web application testing, exploitation, and incident response tasks in this track.

Key learnings

- How to separate internet-facing (NAT) traffic from isolated lab (Host-Only) traffic in VirtualBox
- How to verify VM-to-target connectivity using ping and Nmap before attempting any exploitation
- How to position Burp Suite as an intercepting proxy between a browser and a target application
- How to capture and read live network traffic in Wireshark on a specific virtual interface